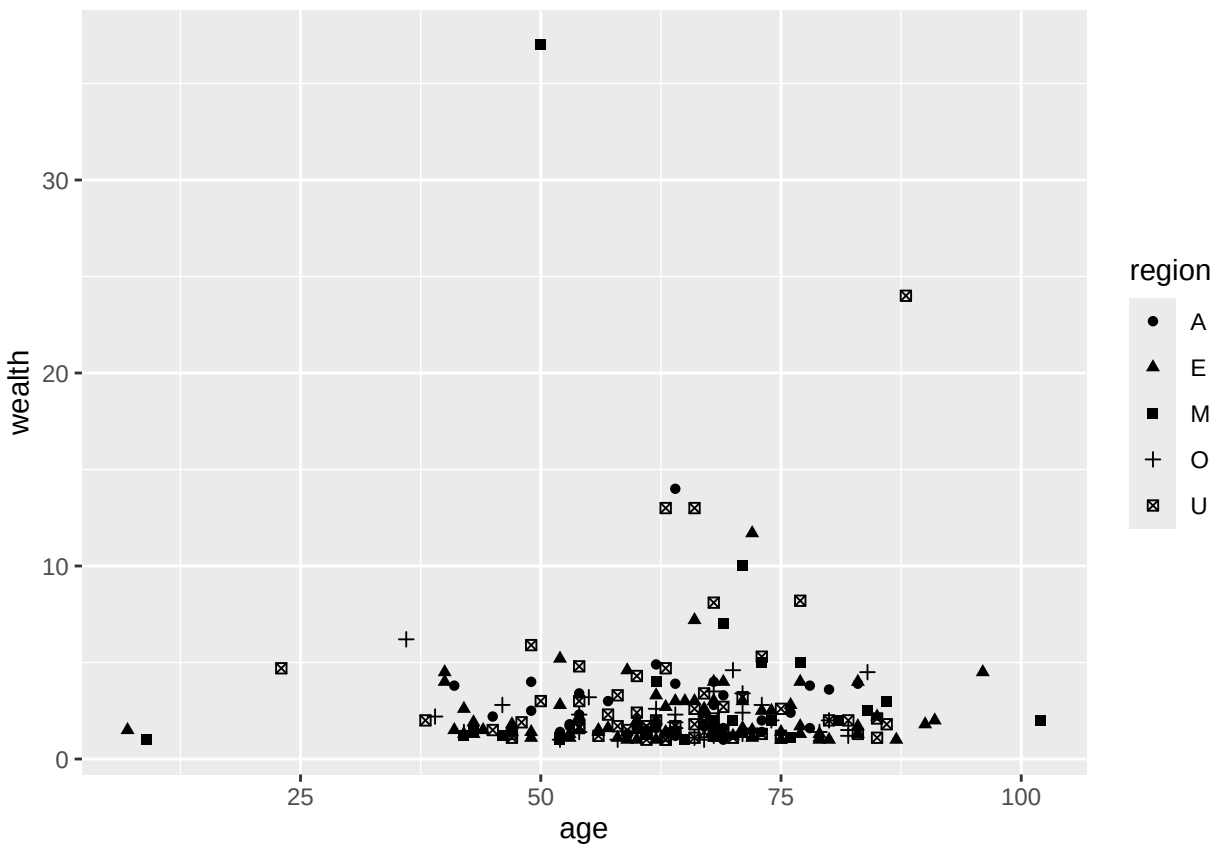
Homework 5

```r
library(faraway)
library(ggplot2)
```

Question 1A

```r
data(fortune)

ggplot(data=fortune) +
  geom_point(aes(x=age, y=wealth, shape=region))
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



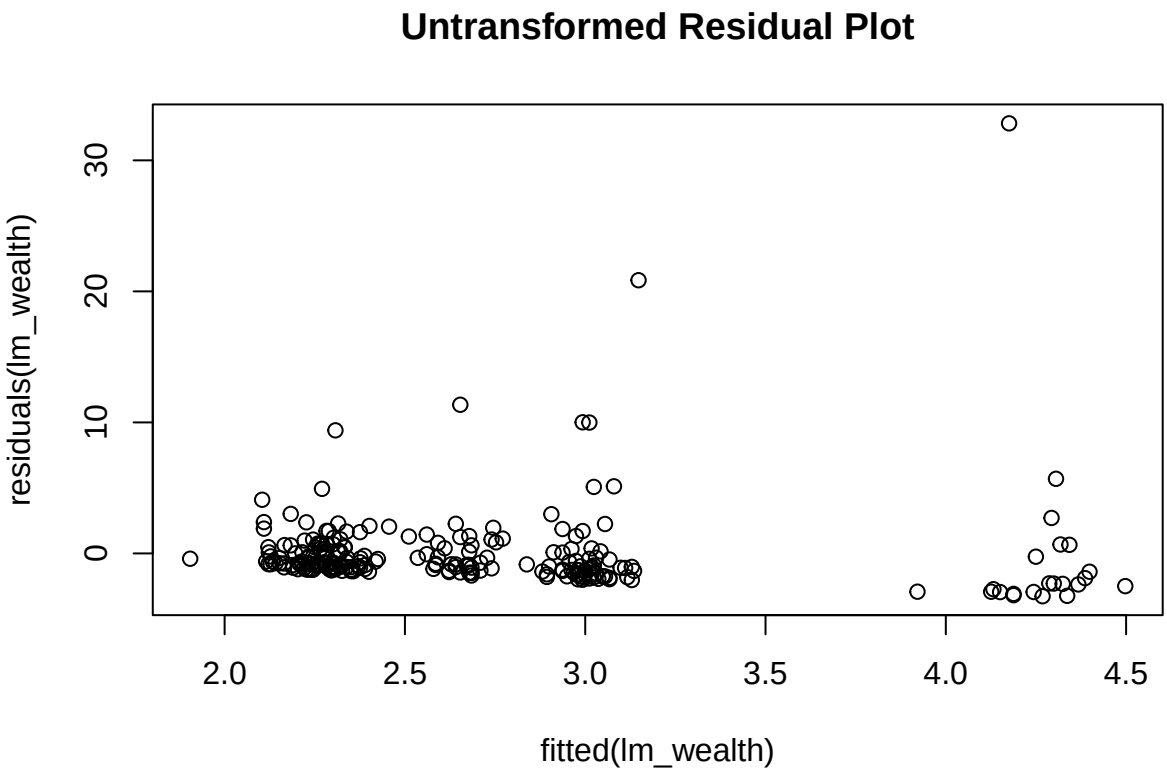
The plot seems to show very little linear correlation between age on wealth. There are also a few very large outliers in the data.

Question 1B

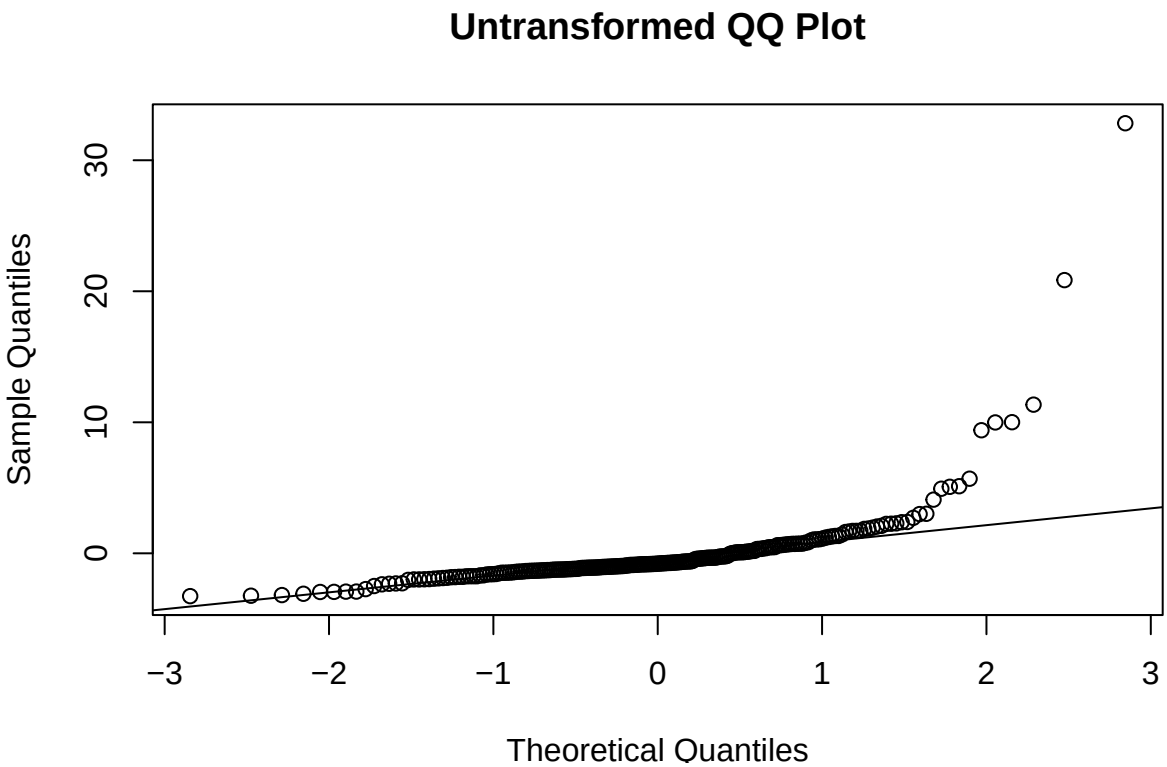
```r
# Untransformed Model
lm_wealth <- lm(I(wealth^1) ~ age + region, data=fortune)
summary(lm_wealth)
```

```
##
## Call:
## lm(formula = I(wealth^1) ~ age + region, data = fortune)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.268 -1.281 -0.785  0.448 32.825
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.256887   1.189041   1.898   0.0590 .
## age          0.006198   0.016554   0.374   0.7085
## regionE     -0.395797   0.671985  -0.589   0.5565
## regionM      1.608699   0.902500   1.782   0.0761 .
## regionO     -0.375843   0.839666  -0.448   0.6549
## regionU      0.345572   0.696402   0.496   0.6202
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.352 on 219 degrees of freedom
##   (7 observations deleted due to missingness)
## Multiple R-squared:  0.03176,    Adjusted R-squared:  0.009652
## F-statistic: 1.437 on 5 and 219 DF,  p-value: 0.2122
```

```r
plot(fitted(lm_wealth), residuals(lm_wealth),
     main='Untransformed Residual Plot')
```



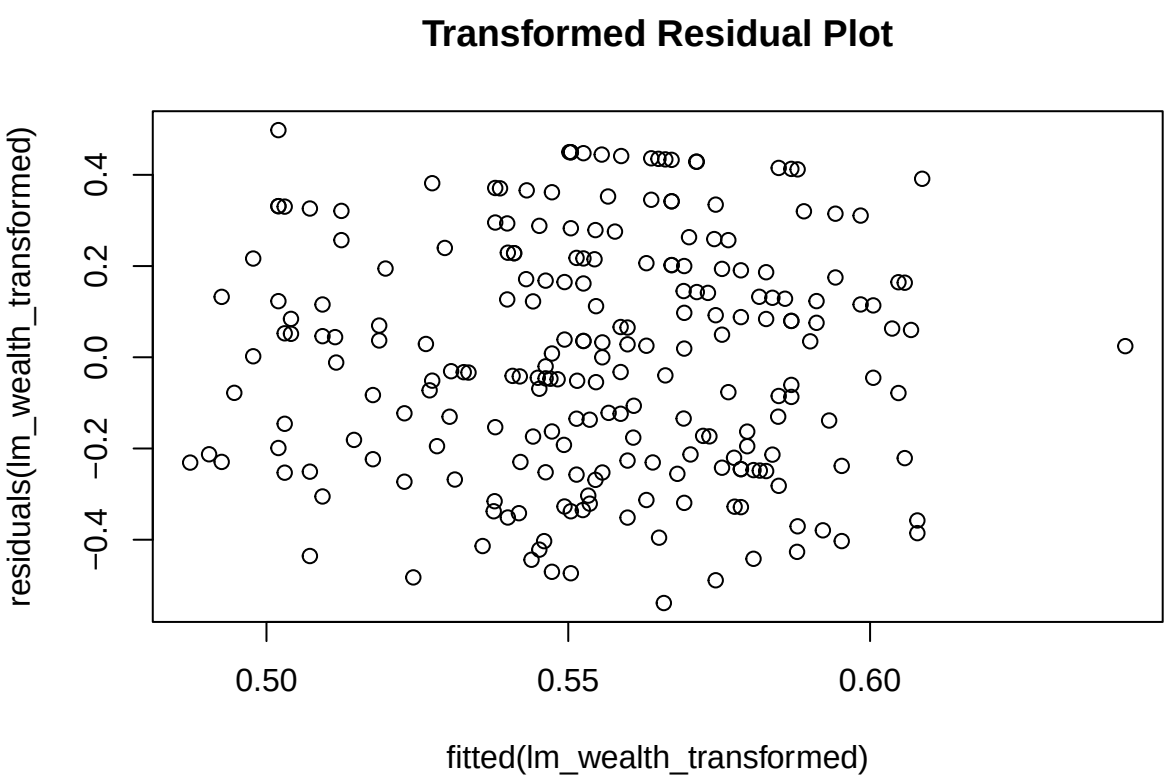

```r
qqnorm(residuals(lm_wealth), main='Untransformed QQ Plot')
qqline(residuals(lm_wealth))
```



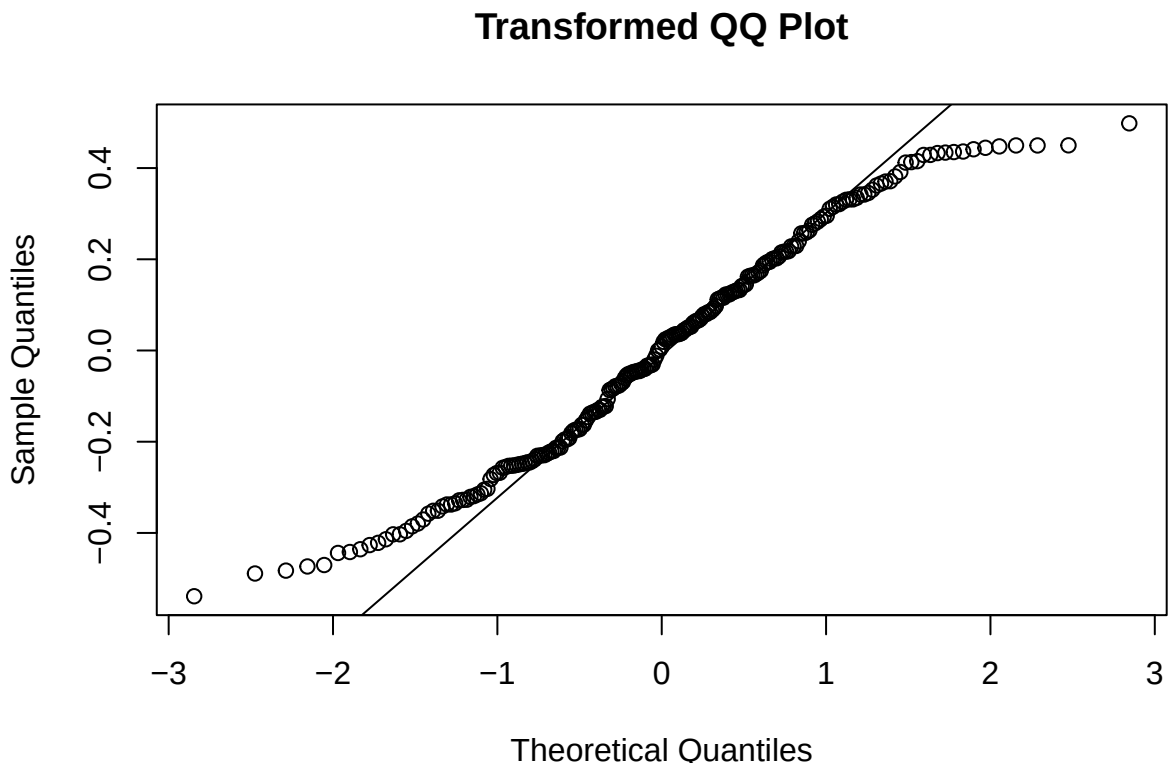

```r
# Transformed Model
lm_wealth_transformed <- lm(I(wealth^-1) ~ age + region, data=fortune)
summary(lm_wealth_transformed)
```

```
##
## Call:
## lm(formula = I(wealth^-1) ~ age + region, data = fortune)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.53881 -0.22113  0.00826  0.20002  0.49803
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.574005   0.091702   6.259 2.01e-09 ***
## age         -0.001044   0.001277  -0.818    0.414
## regionE      0.075583   0.051825   1.458    0.146
## regionM      0.044024   0.069603   0.633    0.528
## regionO      0.051481   0.064757   0.795    0.427
## regionU      0.042186   0.053708   0.785    0.433
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2585 on 219 degrees of freedom
##   (7 observations deleted due to missingness)
## Multiple R-squared:  0.01276,    Adjusted R-squared:  -0.009775
## F-statistic: 0.5663 on 5 and 219 DF,  p-value: 0.7258
```

```r
plot(fitted(lm_wealth_transformed), residuals(lm_wealth_transformed),
     main='Transformed Residual Plot')
```




```r
qqnorm(residuals(lm_wealth_transformed), main='Transformed QQ Plot')
qqline(residuals(lm_wealth_transformed))
```



Question 1C

y-1= 0.574005 - 0.001044xi1+ 0.075583xi2 +0.044024xi3 +0.051481xi4 +0.042186xi5

xi1 - Age xi2 - Europe xi3 - Middle East xi4 - Other xi5 - USA Reference Level: Asia

There doesn't seem to be much of a relationship between age and region on wealth in this dataset. There are no predictors even close to being significant, and the adjusted R-squared is -0.009.

Question 2A

```r
data(pulp)
lm_pulp <- lm(bright ~ operator, data=pulp)
summary(lm_pulp)
```

```
##
## Call:
## lm(formula = bright ~ operator, data = pulp)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -0.440 -0.195 -0.070  0.175  0.560
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  60.2400     0.1458 413.243   <2e-16 ***
## operatorb    -0.1800     0.2062  -0.873   0.3955
## operatorc     0.3800     0.2062   1.843   0.0839 .
## operatord     0.4400     0.2062   2.134   0.0486 *
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.326 on 16 degrees of freedom
## Multiple R-squared:  0.4408, Adjusted R-squared:  0.3359
## F-statistic: 4.204 on 3 and 16 DF,  p-value: 0.02261
```

```r
anova(lm_pulp)
```

```
## Analysis of Variance Table
##
## Response: bright
##           Df Sum Sq Mean Sq F value  Pr(>F)
## operator   3   1.34 0.44667  4.2039 0.02261 *
## Residuals 16   1.70 0.10625
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There appears to be a difference between the operators, as the operator p-value in the one-way ANOVA is significant at the 0.05 level (p value: 0.02261)

Question 2B

```r
pairwise.t.test(pulp$bright, pulp$operator, p.adjust.method='none')
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  pulp$bright and pulp$operator
##
##   a      b      c
## b 0.3955 -      -
## c 0.0839 0.0153 -
## d 0.0486 0.0083 0.7748
##
## P value adjustment method: none
```

```r
pairwise.t.test(pulp$bright, pulp$operator, p.adjust.method='bonferroni')
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  pulp$bright and pulp$operator
##
##   a     b     c
## b 1.000 -     -
## c 0.503 0.092 -
## d 0.292 0.050 1.000
##
## P value adjustment method: bonferroni
```

```r
pairwise.t.test(pulp$bright, pulp$operator, p.adjust.method='fdr')
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
```

```
## data:  pulp$bright and pulp$operator
##
##   a     b     c
## b 0.475 -     -
## c 0.126 0.046 -
## d 0.097 0.046 0.775
##
## P value adjustment method: fdr
```

Pairwise t test without mht correction suggests the difference between level b and d is significant at the 0.01 level (p value: 0.0083. Pairwise t test with bonferroni correction suggest no factor levels are different at the 0.01 level. Same with fdr; however, the p values do get smaller and you see a difference between b and d as well as b and c at the 0.05 level.

Question 3A

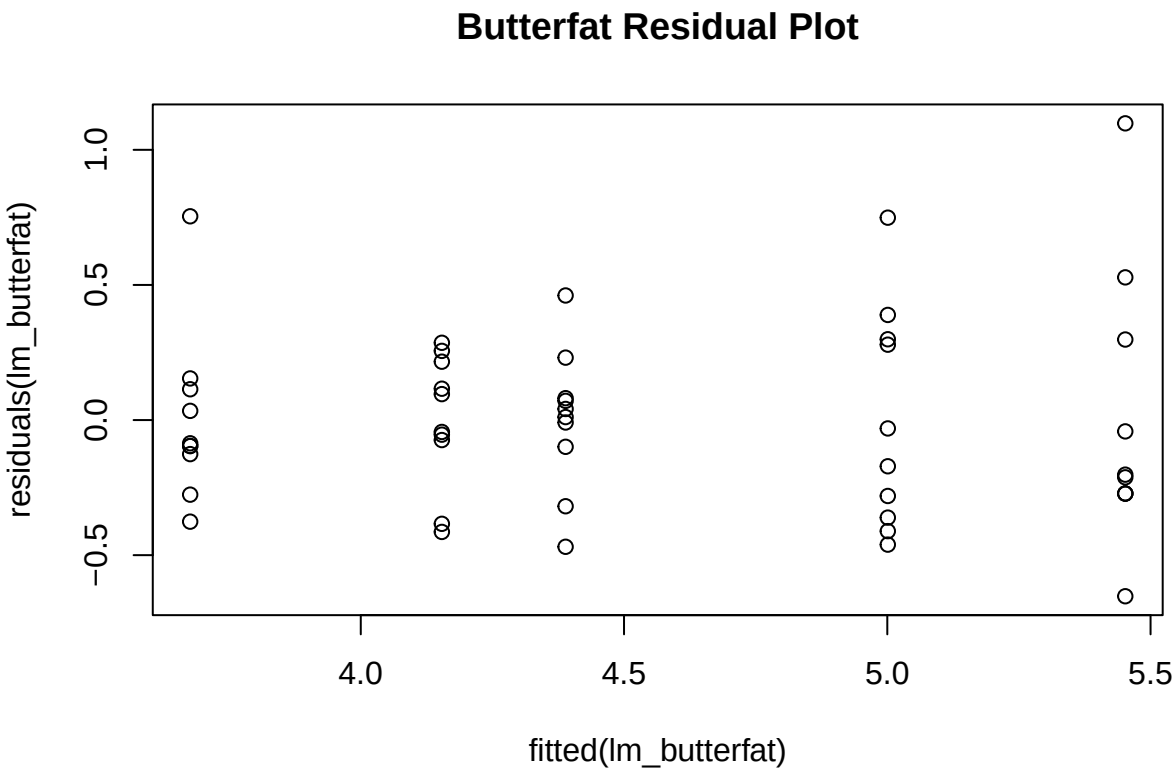
```r
data(butterfat)

mature <- subset(butterfat, Age == 'Mature')
lm_butterfat <- lm(Butterfat ~ Breed, data=mature)
anova(lm_butterfat)
```

```
## Analysis of Variance Table
##
## Response: Butterfat
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## Breed      4 19.6240  4.9060  37.864 7.284e-14 ***
## Residuals 45  5.8306  0.1296
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

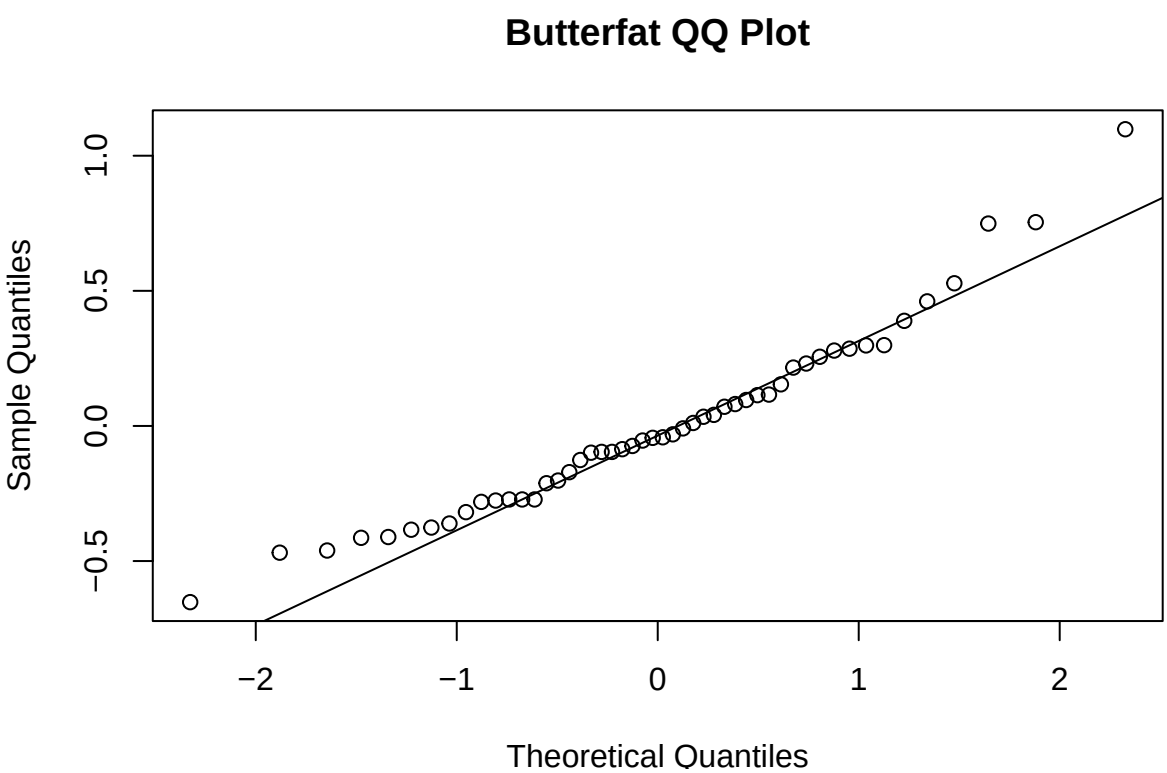
One-way Anova suggests there is a difference between the breeds as the p-value from the f test is highly significant (2e-16).

Question 3B

```r
plot(fitted(lm_butterfat), residuals(lm_butterfat),
     main='Butterfat Residual Plot')
```




```r
qqnorm(residuals(lm_butterfat), main='Butterfat QQ Plot')
qqline(residuals(lm_butterfat))
```



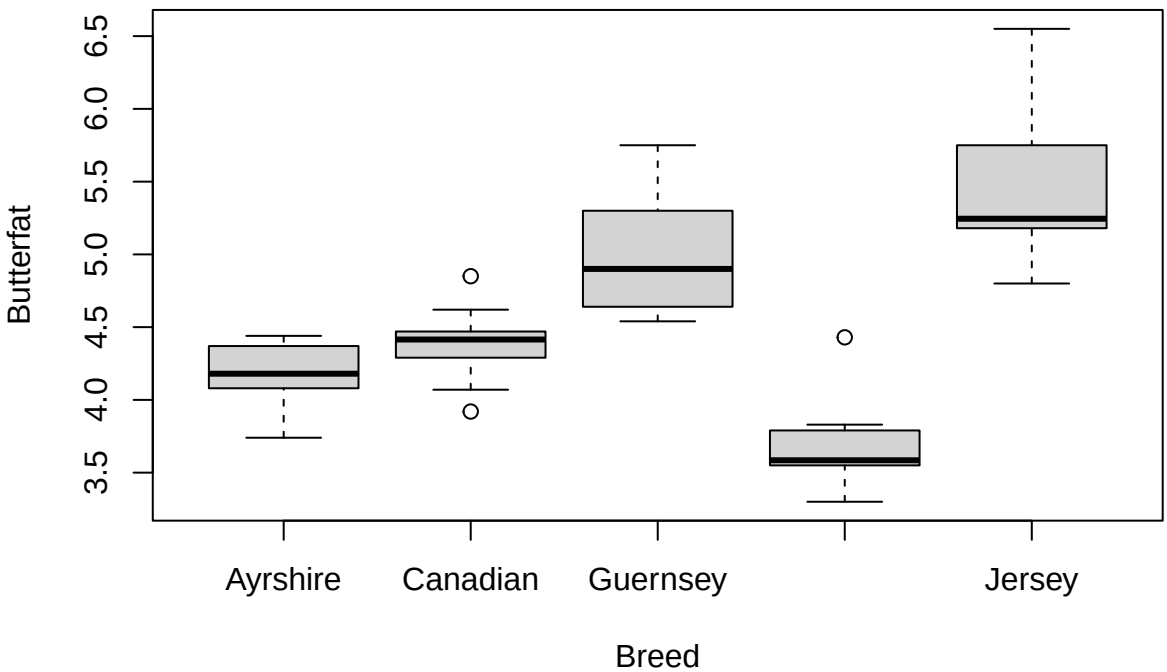

```r
sum(hatvalues(lm_butterfat) > 2*5/50)
```

```
## [1] 0
```

Variance might be a little higher as fitted value increases, but still seems to satisfy our assumptions. Residuals might be slightly long - tailed, but still seems to satisfy our assumptions. There are no large leverage values (no hat values greater than 2p/n).

Question 3C

```
boxplot(Butterfat ~ Breed, data=mature)
```




```
pairwise.t.test(mature$Butterfat, mature$Breed, p.adjust.method='bonferroni')
```

```
##
##  Pairwise comparisons using t tests with pooled SD
##
## data:  mature$Butterfat and mature$Breed
##
##                  Ayrshire Canadian Guernsey Holstein-Fresian
## Canadian         1.0000   -        -        -
## Guernsey         3.8e-05  0.0043   -        -
## Holstein-Fresian 0.0477   0.0006   1.6e-09  -
## Jersey           2.8e-09  4.0e-07  0.0747   2.2e-13
##
## P value adjustment method: bonferroni
```

All breeds are different from each other at the 0.05 level except Guernsey and Jersey as well as Ayrshire and Canada. This appears to be reflected in the boxplot.

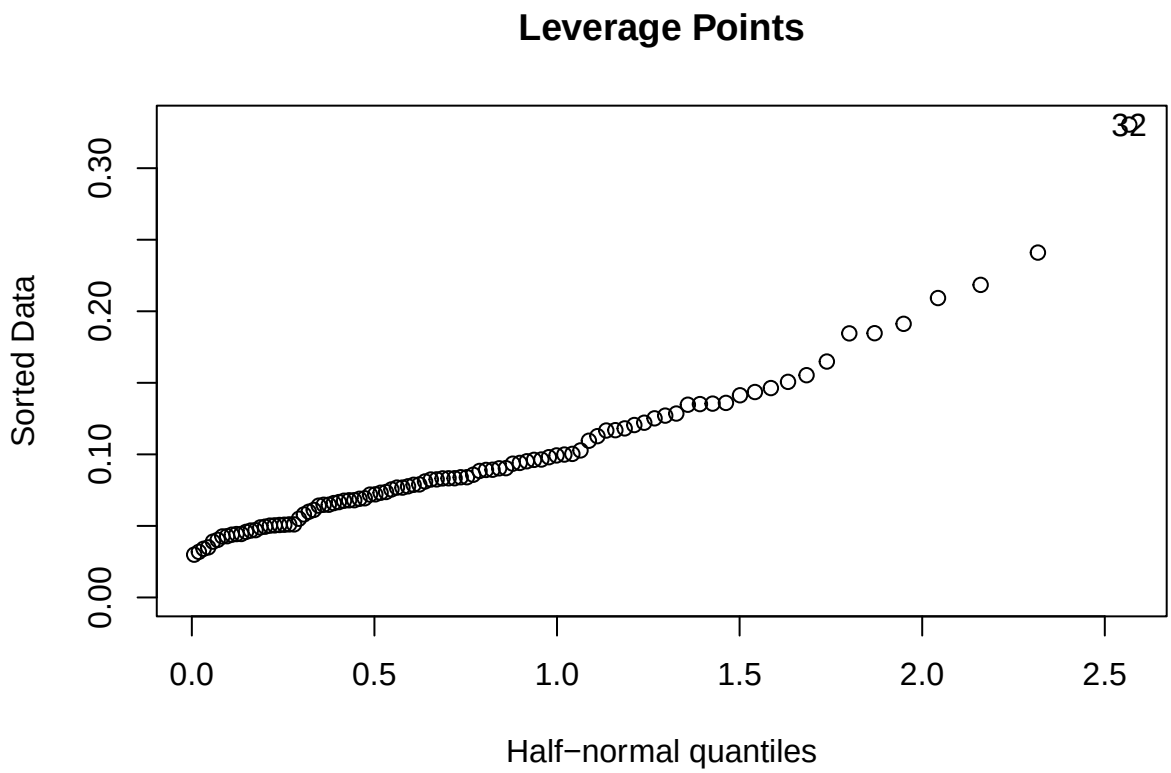
Question 4

```
data(prostate)

lm_prostate <- lm(lpsa ~ lcavol + lweight + age + lbph + svi + lcp +
```

```r
                    gleason + pgg45, data=prostate)

# Checking Leverage Points
sum(hatvalues(lm_prostate) > 2*9/97)
```

```
## [1] 5
```

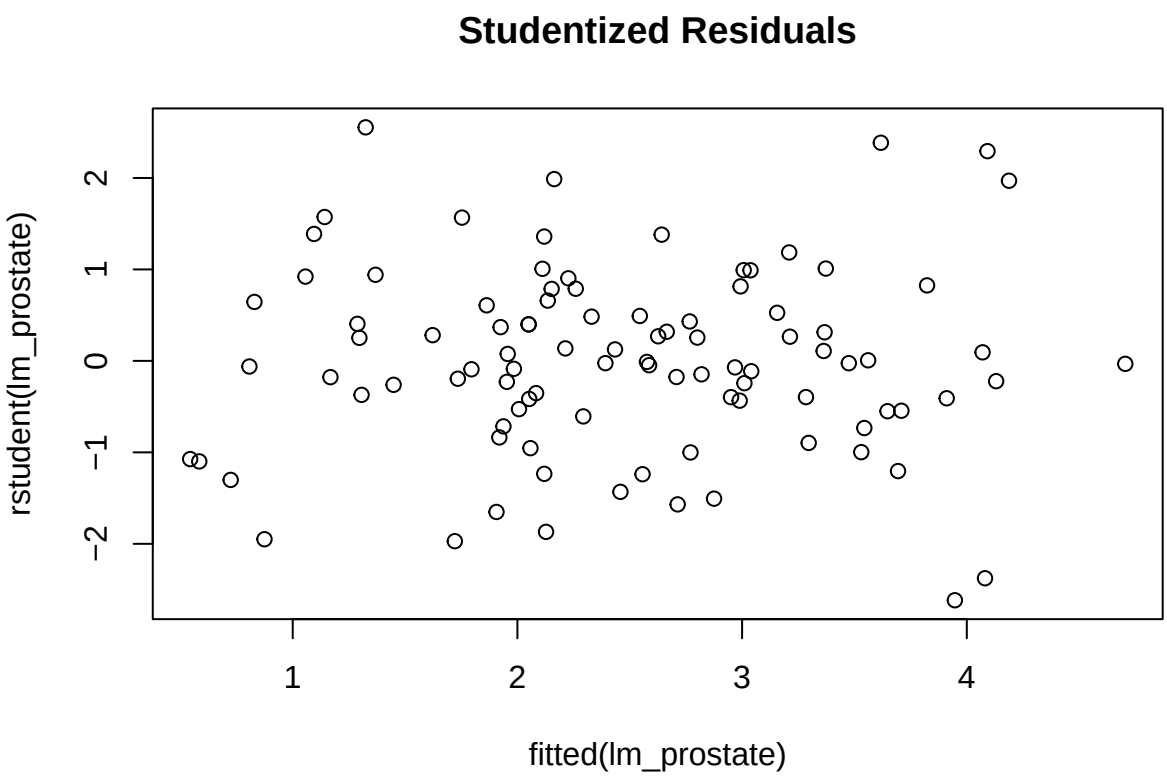
```r
halfnorm(hatvalues(lm_prostate), main='Leverage Points', nlab=0)
```




```r
# Checking Outliers
rstud_pvals <- 2*pt(-abs(rstudent(lm_prostate)), 87)
sum(p.adjust(rstud_pvals, "bonferroni") < 0.05)
```

```
## [1] 0
```

```r
plot(fitted(lm_prostate), rstudent(lm_prostate), main='Studentized Residuals')
```



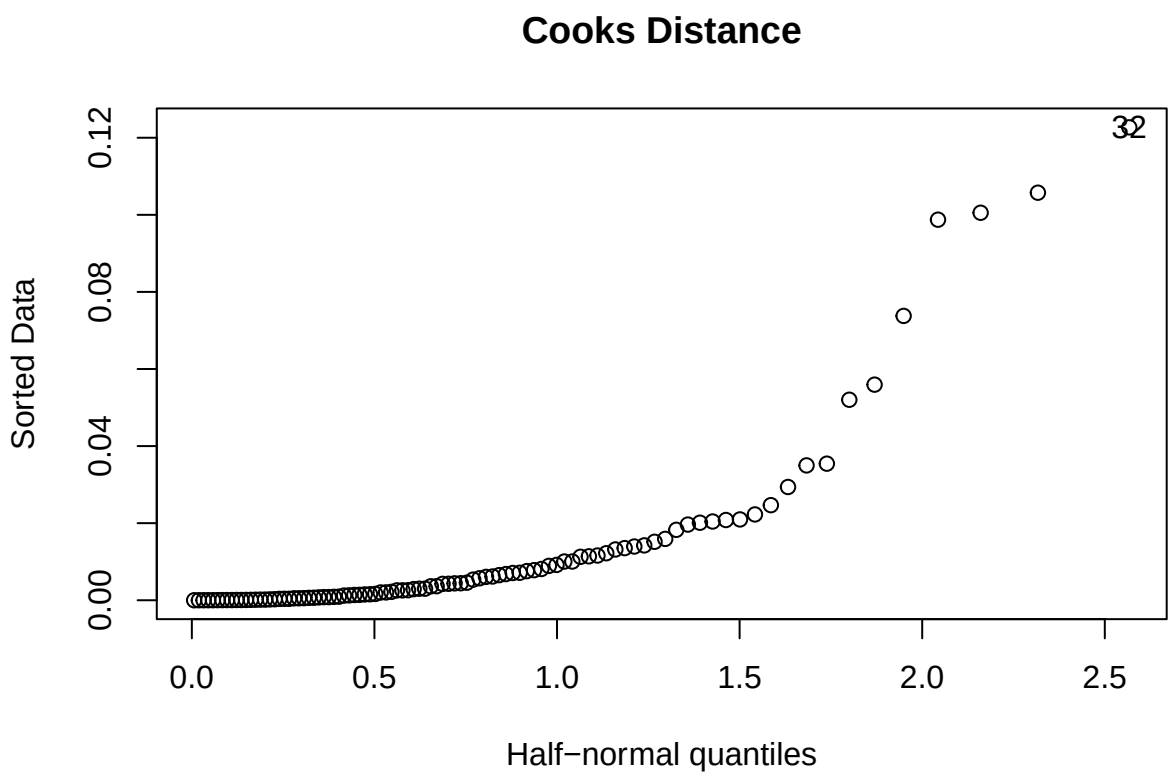

```r
# Uncorrected Test of Outliers
sum(abs(rstudent(lm_prostate)) > 3)
```

```
## [1] 0
```

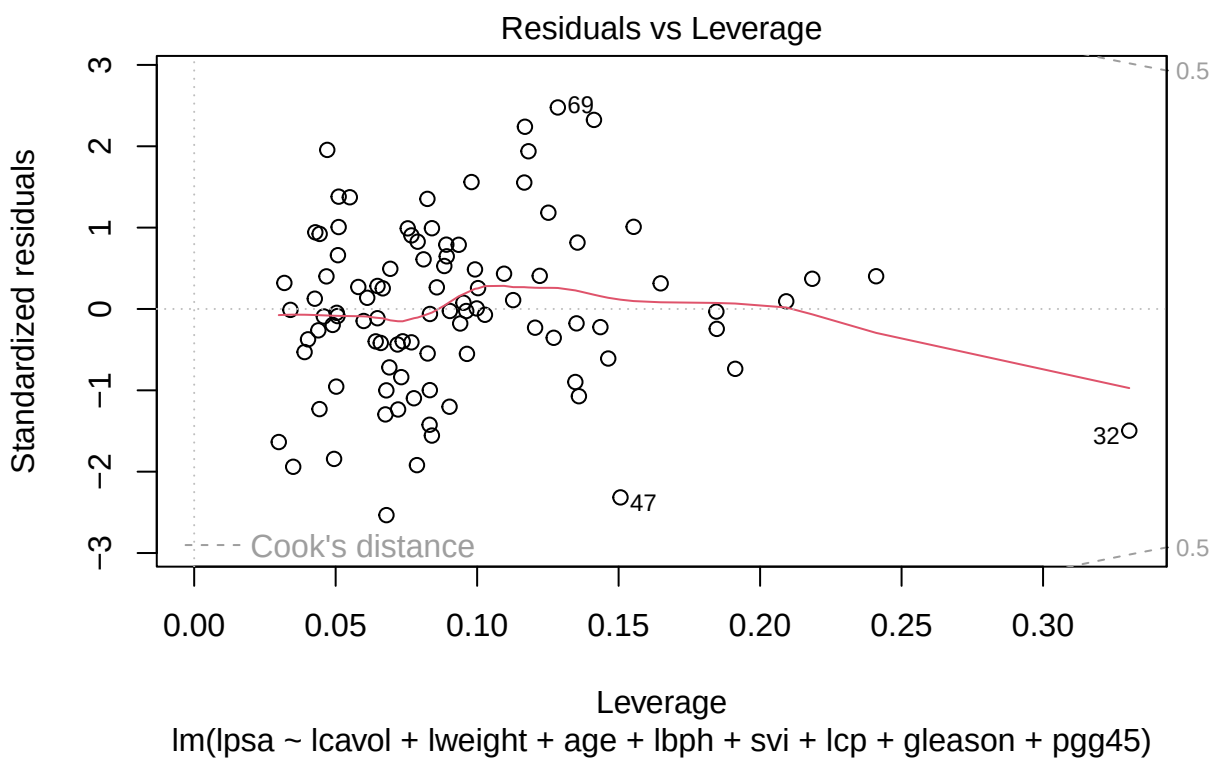
```r
# Checking Influential Points
sum(cooks.distance(lm_prostate) > 4/97)
```

```
## [1] 7
```

```r
halfnorm(cooks.distance(lm_prostate), main='Cooks Distance', nlab=0)
```




```r
plot(lm_prostate, 5)
```



There are five points that could be potential leverage points.Looking at the leverage point plot, there looks to be one clear leverage point, and a few others could be, so five seems to make sense.

Looking at the studentized residual plot , there are no residuals greater than three, so I don't think there are any outliers. I also ran bonferri adjusted test for outliers and found zero.

Looking at the cooks distance graph, there are atleast four points that I would say are influential (maybe as many as 7). Claude said the common heuristic for assessing influential points is checking cooks distance greater than 4/n. This resulted in seven points that could possibly be influential.